

ACU Routing API — IT Deployment & Maintenance Guide

This document contains everything the IT department needs to deploy, maintain, and update the ACU Routing API on a bare-metal Windows Server with minimal friction.

1. Initial Setup (One-Time)

To set up the API on a new Windows Server:

1. **Prerequisites:** Ensure the server has **Python 3.12+** and **PostgreSQL** installed.
2. **Transfer Files:** Copy the entire `centralized_routing_api` folder to the server.
3. **Run the Wizard:** Open **PowerShell as Administrator**, navigate to the project folder, and run:

```
.\scripts\setup_server.ps1
```

4. **Follow Prompts:** The script will automatically ask for the database password and create an initial Administrator account. It will then automatically:
 - Create the `.env` configuration file
 - Install all Python dependencies
 - Safely create the database and required tables
 - Register the API as an auto-restarting Windows Background Service (ACURoutingAPI)

2. General Maintenance

Once the setup wizard finishes, the API runs entirely in the background via Waitress (a multi-threaded production server).

- **Service Name:** ACURoutingAPI
- **Port:** 8080 (Ensure this port is open in the Windows Defender Firewall)
- **Auto-Recovery:** If the API crashes for any reason, Windows will automatically restart it within 5 seconds.
- **Logs:** All access and error logs are automatically written to the `logs\` folder inside the project directory. They rotate daily to prevent disk fill-up.

Stopping/Starting the Service

Open PowerShell as Administrator:

- To stop: `Stop-Service ACURoutingAPI`
- To start: `Start-Service ACURoutingAPI`
- To restart: `Restart-Service ACURoutingAPI`

3. Best Practices for Applying Updates

When the development team provides an update (new code), follow these strict steps to

ensure zero data loss and a clean rollout:

1. **Stop the Service** *Always stop the API before replacing files so files aren't locked.*

```
Stop-Service ACURoutingAPI
```

2. **Replace the Code** Copy the new files over the existing ones. **CRITICAL:** *Do not overwrite the .env file or the logs\ directory.* The .env file contains the server's unique database credentials and secure tokens.

3. **Install Any New Dependencies** If the update includes new libraries, run the dependency installer:

```
python -m pip install -r requirements.txt
```

4. **Restart the Service**

```
Start-Service ACURoutingAPI
```

5. **Verify** Check logs\api_stdout.log to ensure the server started successfully without throwing errors. You can also visit <http://localhost:8080/api/health> to verify liveness.